

Occupational toxicology of asbestos-related malignancies.



INTRODUCTION: Asbestos is banned in most Western countries but related malignancies are still of clinical concern because of their long latencies. This review identifies and addresses some controversial occupational and clinical aspects of asbestos-related malignancies. METHODS: Papers published in English from 1980 to 2009 were retrieved from PubMed. A total of 307 original articles were identified and 159 were included. ASSESSMENT OF EXPOSURE: The retrospective assessment of exposure is usually performed by using questionnaires and job exposure matrices and by careful collection of medical history. In this way crucial information about manufacturing processes and specific jobs can be obtained. In addition, fibers and asbestos bodies are counted in lung tissue, broncho-alveolar lavage, and sputum, but different techniques and interlaboratory variability hamper the interpretation of reported measurements. SCREENING FOR MALIGNANCIES: The effectiveness of low-dose chest CT screening in exposed workers is debatable. Several biomarkers have also been considered to screen individuals at risk for lung cancer and mesothelioma but reliable signatures are still missing. ATTRIBUTION OF LUNG CANCER: Exposures correlating with lung cancer are high and in the same range where asbestosis occurs. However, the unresolved question is whether the presence of fibrosis is a requirement for the attribution of lung cancer to asbestos. The etiology of lung cancer is difficult to define in cases of low-level asbestos exposure and concurrent smoking habits. MESOTHELIOMA: The diagnosis of malignant mesothelioma may also be difficult, because of procedures in sampling, fixation, and processing, and uses of immunohistochemical probes. CONCLUSIONS: Assessment of exposure is crucial and requires accurate medical and occupational histories. Quantitative analysis of asbestos body burden is better performed in digested lung tissues by counting asbestos bodies by light microscopy and/or uncoated fibers by transmission electron microscopy. The benefits of screenings for asbestos-related malignancies are equivocal. The attribution of lung cancer to asbestos exposure is difficult in a clinical setting because of the need to assess asbestos body burden and the fact that virtually all these patients are also tobacco smokers or former smokers. Given the premise that asbestosis is necessary to causally link lung cancer to asbestos, it follows that the assessment of both lung fibrosis and asbestos body burden is necessary.